Theorem (Continuous Functional Calculus).

Let A be a unital C^*-algebra and let $a \in A$ be normal,

$$a^*a = aa^*.$$

Then there exists a unique isometric $*$-homomorphism

$$F_a : C(\sigma_A(a)) \longrightarrow A, \qquad f \longmapsto f(a),$$

such that

$$F_a(\mathrm{id}_{\sigma(a)}) = a, \qquad F_a(1_{\sigma(a)}) = 1_A.$$

Moreover, its image is $C^*(a, 1_A)$.

Statement: Section 2 notes, Theorem 2.22; the image statement is also recorded in the comprehensive study notes, §3.4.

Meaning of "isometric".

$$\|F_a(f)\|_A = \|f\|_\infty = \max_{\lambda \in \sigma_A(a)} |f(\lambda)|.$$

Proof. Step 1: the commutative algebra generated by a.

Set

$$B := C^*(a, 1_A).$$

Because a is normal, a commutes with a^*; hence the $*$-algebra generated by a and 1_A is commutative, and so is its norm closure B.

Tarea 1, Problem 3(d); cf. the comprehensive notes, §4.3, "Un normal genera una álgebra involutiva conmutativa".

By the commutative Gelfand–Naimark theorem,

$$\Lambda_B : B \longrightarrow C(\Omega(B)), \qquad b \longmapsto \widehat{b}, \qquad \widehat{b}(\varphi) = \varphi(b),$$

is an isometric $*$-isomorphism.

Section 2 notes, Theorem 2.20.

Step 2: identify the character space with the spectrum.

By Proposition 2.18,

$$\Phi : \Omega(B) \longrightarrow \sigma_A(a), \qquad \Phi(\varphi) = \varphi(a),$$

is a homeomorphism.

Section 2 notes, Proposition 2.18. Spectral permanence used there is Proposition 2.17; the spectral description by characters ultimately comes from Part 1, Theorem 1.22.

Define

$$\Phi_* : C(\Omega(B)) \longrightarrow C(\sigma_A(a)), \qquad \Phi_*(g) := g \circ \Phi^{-1}.$$

It is plainly linear, multiplicative, unital, and preserves involution. It is isometric because

$$\begin{aligned} \|\Phi_*(g)\|_\infty &= \sup_{\lambda \in \sigma_A(a)} \left|g(\Phi^{-1}(\lambda))\right| \\ &= \sup_{\varphi \in \Omega(B)} |g(\varphi)| = \|g\|_\infty. \end{aligned}$$

It is surjective: if $f \in C(\sigma_A(a))$, set $g = f \circ \Phi$. Then

$$\Phi_*(g) = f \circ \Phi \circ \Phi^{-1} = f.$$

Hence Φ_* is an isometric $*$-isomorphism.

For later use, if $\lambda \in \sigma_A(a)$ and $\varphi = \Phi^{-1}(\lambda)$, then

$$\lambda = \Phi(\varphi) = \varphi(a).$$

Step 3: compute the images of the generators.

Consider the composition

$$\Theta := \Phi_* \circ \Lambda_B : B \longrightarrow C(\sigma_A(a)).$$

Since both factors are isometric $*$-isomorphisms, so is Θ.

Let $\lambda \in \sigma_A(a)$ and $\varphi := \Phi^{-1}(\lambda)$. Then

$$\begin{aligned}\Theta(1_A)(\lambda) &= \Phi_*(\widehat{1_A})(\lambda) = \widehat{1_A}(\Phi^{-1}(\lambda)) \\ &= \widehat{1_A}(\varphi) = \varphi(1_A) = 1 = 1_{\sigma(a)}(\lambda),\end{aligned}$$

so

$$\Theta(1_A) = 1_{\sigma(a)}.$$

Likewise,

$$\begin{aligned}\Theta(a)(\lambda) &= \Phi_*(\widehat{a})(\lambda) = \widehat{a}(\Phi^{-1}(\lambda)) \\ &= \widehat{a}(\varphi) = \varphi(a) = \lambda = \mathrm{id}_{\sigma(a)}(\lambda),\end{aligned}$$

so

$$\Theta(a) = \mathrm{id}_{\sigma(a)}\,.$$

Therefore the inverse

$$F_a := \Theta^{-1} : C(\sigma_A(a)) \longrightarrow B \subseteq A$$

is an isometric $*$-isomorphism and satisfies

$$F_a(\mathrm{id}_{\sigma(a)}) = a, \qquad F_a(1_{\sigma(a)}) = 1_A.$$

This proves existence, and it also shows

$$\mathrm{Im}\, F_a = B = C^*(a, 1_A).$$

Step 4: uniqueness.

Suppose

$$G : C(\sigma_A(a)) \longrightarrow A$$

is another isometric $*$-homomorphism satisfying

$$G(\mathrm{id}_{\sigma(a)}) = a, \qquad G(1_{\sigma(a)}) = 1_A.$$

Because G preserves the involution,

$$G(\overline{z}) = G(z)^* = a^*.$$

Thus G and F_a agree on every polynomial in the coordinate functions z and $\overline{z}$. Those polynomials form a unital $*$-subalgebra of $C(\sigma_A(a))$ that separates points. By the complex Stone–Weierstrass theorem, they are dense in $C(\sigma_A(a))$.

Section 2 notes, Theorem 2.19 (Stone–Weierstrass).

Both G and F_a are continuous (indeed isometric), so equality on the dense subalgebra extends to all of $C(\sigma_A(a))$. Hence

$$G = F_a.$$

This proves uniqueness.

Result map.

- *Normality $\Rightarrow C^*(a, 1_A)$ commutative: Tarea 1, Problem 3(d).*
- *$\Omega(C^*(a, 1_A)) \cong \sigma_A(a)$: Section 2, Proposition 2.18.*
- *Gelfand transform is an isometric $*$-isomorphism: Section 2, Theorem 2.20.*
- *Density of polynomials in $z, \bar{z}$: Section 2, Theorem 2.19.*
- *Continuous functional calculus: Section 2, Theorem 2.22.*

Documents. [P1] Jorge Castillejos, C^*-álgebras: Teoría Espectral en álgebras de Banach; [C2] Jorge Castillejos, C^*-álgebras, Section 2; [T1] C^*-álgebras, Tarea 1; [N] Enrique Díaz Ocampo, Notas de estudio: teoría espectral en álgebras de Banach y C^*-álgebras.